

C++ BPE Tokenizer — Correctness & Latency

Sequence Length	Avg. Latency (μ s)	Median Latency (μ s)	P95 Latency (μ s)	Throughput (tokens/s)	Token-ID Agreement
32	13.456	13.113	13.599	2,378,110.68	100%
64	34.434	34.937	35.881	1,858,625.40	100%
128	80.493	80.536	82.533	1,590,205.33	100%
256	217.708	216.714	231.549	1,175,886.37	100%
512	461.863	461.288	475.437	1,108,553.40	100%
1024	903.909	903.701	925.597	1,132,857.66	100%
2048	1793.720	1791.933	1829.636	1,141,761.08	100%

Table: Correctness and latency characterization of the from-scratch C++ BPE tokenizer.

Token-ID agreement is evaluated against the reference Llama tokenizer for identical input sequences.

Threads	Avg. Latency (μ s)	Throughput (tokens/s)	Parallel Efficiency
1	1838.794	1,113,773.58	100.00%
2	1997.168	2,050,904.35	92.07%
4	1952.213	4,196,262.78	94.19%
8	3127.771	5,238,235.25	58.79%
12	3099.110	7,930,019.49	59.33%
16	3154.586	10,387,415.93	58.29%
20	7129.145	5,745,429.66	25.79%

Table: C++ BPE tokenizer scalability under multi-threaded execution.

Primary Runtime Configuration

Single-threaded execution is used as the primary configuration for batch-size-1 autoregressive inference.

Aggregate throughput scales from 1.11M tokens/s on one thread to 10.39M tokens/s at 16 threads. Beyond this point, contention becomes significant, with throughput falling to 5.75M tokens/s at 20 threads.

C++ BPE Tokenizer — Latency Breakdown & Memory

Component	Latency (μ s)	% of Total
Text Preprocessing / Regex Splitting	90.735	4.031%
Byte-Level Conversion	59.750	2.655%
BPE Merge Processing	1837.448	81.639%
Vocabulary Lookup + Token ID Generation	96.209	4.275%
Total Tokenization	2250.710	100.000%

Benchmark Configuration: 2048-token target, 2081 actual tokens, 100 iterations, 20 warmups
BPE merging is the dominant bottleneck, consuming 81.64% of total tokenizer latency.

Average Latency: **1.082 μ s/token** Throughput: **924,597 tokens/s**

Memory Footprint

Peak RSS: 55.61 MB